AP BIOLOGY TOPIC 4.6

Regulation of Cell Cycle: Checkpoints, Cyclin–CDKs, p53, Oncogenes & Cancer

UNIT 4: CELL COMMUNICATION & CELL CYCLE (10–15% EXAM WEIGHT)

IST-1.D | IST-1.E | CYCLINS / CDKS & TUMOR SUPPRESSORS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 4.6 (REGULATION OF CELL CYCLE: CHECKPOINTS, CYCLIN–CDKS, P53, ONCOGENES & CANCER)

CED Core Vocabulary

Purpose of Checkpoints — Internal molecular surveillance mechanisms that pause the cell cycle to verify fidelity and prevent damaged or unprepared cells from dividing. Ensure stages...

Cyclins — Regulatory proteins whose concentrations oscillate (rise and fall cyclically) across the cell cycle as they are synthesized and degraded.

Phosphorylation of Targets — Active CDK uses ATP to phosphorylate target proteins (e.g., nuclear lamins to fragment the nuclear envelope, condensins to condense chromosomes, spindle...

External Regulatory Signals in Normal Cells — Growth Factors (e.g., PDGF): Paracrine ligands secreted by platelets/cells that bind RTK receptors to stimulate target cells to pass the G_1 checkpoint and divide.

Cyclin-Dependent Kinases (CDKs) — Enzymes present at a constant concentration throughout the entire cell cycle, but they are catalytically INACTIVE on their own .

Synthesis & Binding — As a cell grows in S/G_2 , mitotic cyclin accumulates and binds allosterically to CDK → forms the active Cyclin–CDK complex (MPF) .

Ubiquitin-Proteasome Degradation — In late mitosis (Anaphase), the cyclin subunit is enzymatically degraded → CDK becomes inactive again → cell exits mitosis into G_1 .

Anchorage Dependence — Normal animal cells must be physically attached to an extracellular matrix (ECM) substratum via integrins in order to divide.

§2. THE 3 MAJOR INTERNAL CELL CYCLE CHECKPOINTS

Quality Control Gates

- **Purpose of Checkpoints:** Internal molecular surveillance mechanisms that pause the cell cycle to verify fidelity and prevent damaged or unprepared cells from dividing. Ensure stages occur **in the correct order, at the right time, and only once per cycle**.

CHECKPOINT	TIMING	WHAT THE CELL INSPECTS	IF CHECKPOINT FAILS
1. G_1 Checkpoint *(Restriction Point)*	Near end of G_1 (**BEFORE** DNA replication)	Adequate **cell size**, sufficient **nutrients/ATP**, external **growth factors**, and **undamaged DNA**.	Pauses cycle for DNA repair, exits into non-dividing G_0 **phase**, or undergoes **apoptosis**.
2. G_2 Checkpoint	End of G_2 (**AFTER** S-phase replication, before Mitosis)	Verifies **DNA replication is 100% complete**, newly replicated **DNA is undamaged**, and MPF levels are sufficient.	Arrests at G_2 to activate DNA repair enzymes; if irreparable → triggers **apoptosis**.
3. M (Spindle) Checkpoint	During **Metaphase** (before Anaphase)	Ensures **ALL chromosomes' kinetochores are properly attached** to spindle microtubules from opposite poles.	Blocks separase activation (delays **Anaphase**) to prevent **nondisjunction / aneuploidy**!

[COMMON AP EXAM FRQ TRAP] Two High-Yield Checkpoint Mistakes:

1. G_1 **checks DNA BEFORE replication** (to avoid copying mutated templates), whereas G_2 **checks DNA AFTER replication** (before mitosis begins).
2. At the **M checkpoint**, if even a single chromosome is unattached to spindle fibers, the cell **must NOT proceed from Metaphase to Anaphase**!

§4. EXTERNAL SIGNALS, CANCER & LOSS OF CHECKPOINT CONTROL

Uncontrolled Division

- **External Regulatory Signals in Normal Cells:**
  - **Growth Factors (e.g., PDGF):** Paracrine ligands secreted by platelets/cells that bind RTK receptors to stimulate target cells to pass the G_1 checkpoint and divide.
  - **Density-Dependent Inhibition (Contact Inhibition):** Crowded cells touch one another and stop dividing once a single confluent monolayer fills a surface.
  - **Anchorage Dependence:** Normal animal cells must be physically attached to an extracellular matrix (ECM) substratum via integrins in order to divide.
- **Hallmarks of Cancer Cells (Failure of Cell Cycle Regulation):**
  - **Ignore internal checkpoints** (G_1, G_2, M) and continue dividing even with severe DNA damage.
  - Make their own growth factors (autocrine loop) or have **constitutively active receptors/kinases**.
  - Lose **density-dependent inhibition** (pile up into tumors) and lose **anchorage dependence** (metastasize through blood/lymph).
  - **Evade apoptosis** and express *telomerase* for replicative immortality.

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§3. MOLECULAR ENGINE: CYCLINS & CYCLIN-DEPENDENT KINASES (CDKS)

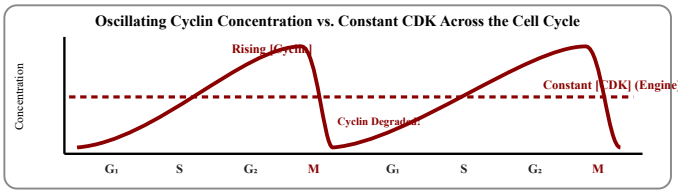
CDK = Engine | Cyclin = Key

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Molecular Engine ➔ **Cyclin-Dependent Kinases (CDKs)** ➔ **Cyclins**

- **Cyclin-Dependent Kinases (CDKs):**
  - Enzymes present at a **constant concentration** throughout the entire cell cycle, but they are **catalytically INACTIVE on their own**.
- **Cyclins:**
  - Regulatory proteins whose concentrations **oscillate (rise and fall cyclically)** across the cell cycle as they are synthesized and degraded.
- **How Cyclin–CDK Control Works (e.g., MPF — Maturation Promoting Factor):**

1. **Synthesis & Binding:** As a cell grows in **S/G₂**, mitotic **cyclin accumulates** and binds allosterically to **CDK** → forms the active **Cyclin–CDK complex (MPF)**.
2. **Phosphorylation of Targets:** Active CDK uses ATP to **phosphorylate target proteins** (e.g., nuclear lamins to fragment the nuclear envelope, condensins to condense chromosomes, spindle proteins) → pushes cell past the **G₂** checkpoint into **Mitosis**.
3. **Ubiquitin-Proteasome Degradation:** In late mitosis (Anaphase), the **cyclin subunit is enzymatically degraded** → CDK becomes inactive again → cell exits mitosis into **G₁**.



§5. PROTO-ONCOGENES VS. TUMOR SUPPRESSOR GENES (P53)

Gas Pedal vs Brakes

GENE CLASS	NORMAL CELLULAR FUNCTION	CANCER-CAUSING MUTATION & ANALOGY
Proto-Oncogenes → Oncogenes *(e.g., Ras, Myc, HER2)*	Normally **promote cell growth and division** in a regulated way in response to growth factors.	**Gain-of-Function (Dominant — 1 mutant allele needed):** Converts proto-oncogene into an overactive **oncogene**. **Analogy = "Gas pedal stuck down!"**
Tumor Suppressor Genes *(e.g., p53, BRCA1/2, Rb)*	Normally **inhibit cell division, repair damaged DNA, or trigger apoptosis** when DNA is irreparable.	**Loss-of-Function (Recessive — 2 mutant alleles needed):** Removes normal checkpoint inhibition. **Analogy = "Cut the brakes!"**

- **Spotlight on p53 ("Guardian of the Genome" — Mutated in >50% of Human Cancers):**
  - When DNA damage (UV, radiation, replication error) is detected at the **G₁** checkpoint, the **p53 transcription factor** is activated and performs **3 protective actions**:
    1. **Halts the Cell Cycle at G₁:** Transcribes *p21*, a CDK-inhibitor protein that blocks Cyclin–CDK so the cell has time to repair its DNA.
    2. **Activates DNA Repair Enzymes** to fix the lesion.
    3. **Triggers Apoptosis (Programmed Cell Death):** If DNA damage is too severe to repair, p53 activates caspase apoptosis genes so the mutated cell destroys itself.
  - **Consequence of Defective/Mutated p53:** Cells with damaged DNA fail to pause or undergo apoptosis → **survive, replicate damaged DNA, and accumulate mutations → cancer!**

[MECHANISTIC NOTE] **AP Exam Scope Note:** You do NOT need to memorize the names of specific cyclin-CDK pairs (like Cyclin B / CDK1) or specific growth factors — focus on how **cyclin accumulation/degradation regulates CDK activity** and how **checkpoint or p53 mutations lead to cancer or apoptosis failure**.

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 4.6 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 4.6: Regulation of Cell Cycle: Checkpoints, Cyclin–CDKs, p53, Oncogenes & Cancer ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 (Mitosis in Onion Root Tips & Lectin Signaling): Root tip squash stained with aceto-orcein/toluidine blue to compare mitotic index (% dividing cells = [cells in PMAT / total cells counted] × 100) between control meristem and phytohemagglutinin (lectin) or caffeine-treated roots. Chi-square (χ^2) tests whether observed mitotic vs. interphase counts differ significantly from control. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Mitotic Index = (N_mitosis / N_total) × 100. DNA Content (C) & Ploidy (n) tracking in diploid (2n) somatic cells: G_1 = 2n, 2C (unreplicated chromosomes); S phase = DNA synthesis (2n, 2C → 4C); G_2/Prophase/Metaphase = 2n, 4C (sister chromatids joined); Anaphase = transient 4n, 4C (chromatids separate into individual chromosomes); Post-Cytokinesis = 2n, 2C. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Signaling Disruption Case Study: Vibrio cholerae secretes Cholera Toxin, which enzymatically modifies the G_α subunit of intestinal GPCRs so it cannot hydrolyze GTP → GDP. Locked in its active GTP-bound state, G_α continuously activates adenylyl cyclase, causing runaway cAMP accumulation, constitutive PKA activation, CFTR Cl^-/H_2O efflux, and life-threatening secretory diarrhea. In cell cycle control, loss-of-function mutations in both alleles of TP53 ('guardian of the genome') prevent p21-mediated G_1/S arrest and apoptosis of DNA-damaged cells, driving oncogenic proliferation.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 4.6 to prove why the claim holds.

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